

A. BABAEI

Tehran, Iran
+98 919 651 5051
ablofazlbabaei@gmail.com
[Portfolio](#) | [GitHub](#)

Biomedical Engineer | Neuroscience Researcher | Prospective PhD Applicant

Translational neuroscience · Neural engineering · Neuromodulation

PROFILE

Biomedical engineer with an M.Sc. in Biomedical Engineering (Bioelectric) and hands-on experience in preclinical neuroscience, neuromodulation, rodent behavioral research, scientific programming, biomedical instrumentation, and technical scientific communication. Seeking doctoral training that integrates translational neuroscience, neurocognitive systems, neural engineering, and clinically relevant neurotechnology.

RESEARCH INTERESTS

- Translational neuroscience
- Deep brain stimulation and circuit neuromodulation
- Biomedical device design and development
- Rodent neurosurgery and preclinical disease models
- Brain-behavior relationships and quantitative phenotyping
- Neurocognitive neuroscience
- Neurodegenerative diseases and disease-modifying or restorative approaches
- Rehabilitation engineering and assistive or rehabilitative devices
- Neural signal acquisition, instrumentation, and analysis
- Scientific computing and reproducible biomedical analysis

EDUCATION

M.Sc. in Biomedical Engineering (Bioelectric) | Iran University of Science and Technology
Tehran, Iran

2023-2026

- Graduate focus: neural engineering, neuromodulation, machine learning, biomedical signal processing, and image processing.
- Research integrated preclinical experimentation, deep brain stimulation, behavioral assessment, and computational analysis.

B.Sc. in Biomedical Engineering | Islamic Azad University, Tehran East
Tehran, Iran

2019-2022

- GPA: 18.68/20.
- Bachelor's project: Design and Development of a Custom-Built Neural Potential Recording Device.
- Contributed to the launch of the university scientific journal and participated in research activities.

RESEARCH EXPERIENCE

Graduate Researcher - Neuromodulation and Behavioral Neuroscience | Master's Research Project
Tehran, Iran

2023-2026

- Investigated parafascicular thalamic stimulation in a unilateral 6-OHDA rat model of Parkinsonian motor dysfunction.
- Performed or supported rodent handling, stereotaxic procedures, lesion validation, electrode-implantation workflows, deep brain stimulation, and behavioral testing.
- Developed video-based behavioral-analysis workflows and customized computational pipelines for event extraction and structured datasets.
- Applied repeated-measures statistical methods, model comparison, dose-response analysis, multiple-testing control, and publication-oriented visualization in Python.
- Prepared manuscript sections, supplementary tables, scientific figures, graphical abstracts, and submission documentation.

Biomedical Instrumentation Project | Bachelor's Capstone Project
Tehran, Iran

2021-2022

- Designed and developed a custom-built device for recording neural potentials.
- Integrated biomedical instrumentation principles, signal acquisition, circuit-level development, and system testing.

METHODS AND TECHNICAL SKILLS

- Rodent handling and behavioral testing
- 6-OHDA disease models and DBS protocols
- GEE, mixed models, model comparison, and FDR control
- Image processing and machine learning
- Scientific visualization and reproducible reporting
- Stereotaxic and rodent neurosurgical workflows
- Python and MATLAB
- Neural and biomedical signal processing
- Video-based behavioral analysis
- Manuscript, figure, and graphical-abstract preparation

TEACHING AND SCIENTIFIC COMMUNICATION

Instructor - Practical MATLAB for Neuroscience | Amirkabir University of Technology,
International Neuroscience School
Tehran, Iran

Nov 2025

- Designed and delivered six practical sessions (12 hours) focused on MATLAB applications in neuroscience.

Course Developer and Instructor - MATLAB Beginner to Advanced | Tech Academy
Tehran, Iran

Dec 2023-Present

- Produced a structured 40-hour offline course comprising 38 sessions, including lesson plans, code examples, presentations, exercises, and neuroscience-oriented applications.

RESEARCH OUTPUTS AND PROFESSIONAL LINKS

Peer-reviewed outputs: Four articles reported in the source CV (two ISI-indexed and two ISC-indexed). Complete bibliographic citations should be inserted before formal submission.

Code and portfolio: github.com/A-Babaei | a-babaei.github.io

RELEVANT TECHNICAL EXPERIENCE

Medical Equipment Supervisor Engineer (Part-Time) | Peyramoon Teb Co.
Tehran, Iran

Dec 2023-Mar 2026

- Conducted device inspections, compliance checks, technical assessments, installation support, maintenance coordination, and troubleshooting.

Medical Equipment Repair Technician | Vistatech-Pishro & Vista Co.
Tehran, Iran

Mar 2023-May 2023

- Worked with sterilization systems, autoclaves, ultrasonic devices, dental handpieces, and basic laboratory equipment.